Controlled Markov game vs. the LLM naming game

Theoretical companion — what changes with more than two options

05.08.2026 • companion to *Controlled Markov game: full mathematical walkthrough*

The walkthrough defined our controlled game and computed its conditional mutual information. This companion asks two questions the walkthrough did not: *how does that model differ, as a mathematical object, from the naming game of Ashery, Aiello and Baronchelli*, and *what genuinely changes when the name pool grows past two*. The estimation statistics come last, in §7, because they only make sense once the geometry is clear.

Reference for the empirical model throughout: Ashery, Aiello & Baronchelli, *Emergent social conventions and collective bias in LLM populations*, Science Advances **11**(20) eadu9368 (2025); preprint arXiv:2410.08948.

1 The two models as mathematical objects

Write $A = \{1, \dots, W\}$ for the name pool, N for the population size.

	Ours	**Theirs**
Name pool	W, free	$W = 10$; $W = 2$ for the bias analysis; up to 26
Population	$N \le 24$	$N = 24$, up to 200
Who acts per step	all N (synchronous)	one random pair (asynchronous)
Agent memory	current partner's name only	last $H = 5$ interactions
Memory contents	—	own name, partner's name, success, running score
Payoff channel	none	reward on match, penalty on mismatch
Decision rule	explicit stochastic kernel	LLM forward pass, samplable only
Markov in the action profile	**yes**	**no** — hidden Markov
Name-permutation symmetry	exact	broken (§4.3)

The last two rows are the substantive ones; everything else is parametric. They are developed in §2.4 and §4.3.

2 Operators

2.1 Transfer and Koopman

For a controlled chain with input u, the transfer (Frobenius–Perron) operator acts on measures,

$$(\mathcal{T}_u^* \mu)(s') = \sum_s \mu(s)\, T_u[s, s'],$$

and its adjoint, the Koopman operator, acts on observables,

$$(\mathcal{K}_u f)(s) = \sum_{s'} T_u[s,s'] f(s') = \mathbb{E}[f(s_{t+1}) \mid s_t = s, u].$$

Empowerment is naturally a Koopman statement: it compares the one-step-controlled propagator $\mathcal{K}_u \bar{\mathcal{K}}^{h-1}$ against the uncontrolled $\bar{\mathcal{K}}^h$.

2.2 Projection and lift

Coarse-graining by $\varphi : \mathcal{S} \to \mathcal{M}$ gives the projection $\Pi = \Phi^\top$, $(\Pi\mu)(m) = \sum_{s\in\varphi^{-1}(m)} \mu(s)$, whose fibres $\varphi^{-1}(m)$ are the sets of microstates sharing a macrostate. A *lift* Λ_d needs a reference measure d to decide how mass sits inside each fibre,

$$(\Lambda_d \nu)(s) = \nu(\varphi(s)) \cdot \frac{d(s)}{\sum_{s''\in\varphi^{-1}(\varphi(s))} d(s'')}.$$

Always $\Pi\Lambda_d = \mathrm{Id}$; never $\Lambda_d\Pi = \mathrm{Id}$. That asymmetry is the information destroyed by coarse-graining, and §7 is about paying for it.

2.3 Lumpability

$\{m_t\}$ is Markov for every initial condition iff the block row-sums $\sum_{s'\in\varphi^{-1}(m')} T[s,s']$ are constant over $s \in \varphi^{-1}(m)$ (Kemeny–Snell; operator form $\Pi\mathcal{T} = \Pi\mathcal{T}\Lambda\Pi$ for every lift). Otherwise the observed process is a *hidden Markov* process and the correct h-step object is the composition lift → propagate → project,

$$\mathcal{G}_u^{(h)} = \Pi D_{w_u} \mathcal{T}_u \bar{\mathcal{T}}^{h-1} \Pi^*,$$

which is the $G_u = \Phi^\top \mathrm{diag}(w_u) R_u \Phi$ of the walkthrough.

2.4 Memory augmentation, and why it matters

Their agents carry a buffer of the last H interactions. Success and running score are functions of that buffer, so the state is

$$\tilde{\mathcal{S}} = (A^{2H})^N, \qquad |\tilde{\mathcal{S}}| = W^{2HN},$$

and the kernel is

$$\tilde{T}[\tilde{s},\tilde{s}'] = \sum_{\{i,j\}} P(i,j) \sum_{c_i,c_j\in A} \pi_{\mathrm{LM}}(c_i \mid \mu_i)\, \pi_{\mathrm{LM}}(c_j \mid \mu_j)\, \mathbf{1}[\tilde{s}' = \mathrm{shift}(\tilde{s}, i, j, c_i, c_j)],$$

with μ_i agent i's buffer and shift the deterministic push-and-drop.

This *is* a Markov chain — memory augmentation always restores the Markov property. The point is what it costs. At their defaults $W = 10$, $H = 5$, $N = 24$ the state space has 10^{240} elements, and π_{LM} has no closed form. Both facts are permanent.

> **The single sharpest structural difference.** For us the observable *is* the state: the action profile is Markov. For them the observable is a function of a much larger hidden state, so the observed process is a hidden Markov process. Consequently M_t is not a sufficient statistic for the future, and
>
> $$I(M_{t+h}\,;\,\mathrm{past} \mid M_t) > 0.$$
>
> Conditioning on any *observable* macrostate cannot close that gap, because the memory

> buffers that carry the dependence are never observed.

3 Geometry of the W-name state space

3.1 From a segment to a simplex

Exchangeable agents mean the sufficient statistic is the composition $n = (n_1, \dots, n_W)$ with $n_c = \#\{i : a_i = c\}$ and $\sum_c n_c = N$. These are the lattice points of the scaled simplex $N\Delta^{W-1}$, and there are $\binom{N+W-1}{W-1}$ of them.

W	macrostate is	dimension	lattice points, $N = 24$
2	a segment	1	25
3	a triangle	2	325
4	a tetrahedron	3	2 925

> The order parameter stops being a scalar. At W names the macroscopic description is $(W-1)$-dimensional. This is a qualitative change, not a bigger version of the same thing — and it is the origin of every difficulty in §7.

3.2 Consensus is an orbit, not two points

The absorbing states are the W vertices Ne_c. At $W = 2$ they are the two endpoints of a segment and the dynamics has a single reaction coordinate. At $W \geq 3$ the interior carries more structure: the barycentre $(N/W, \dots, N/W)$ is a symmetric fixed point of the mean-field flow, and the faces of the simplex are invariant sets corresponding to *partial consensus* — one or more names already extinct.

Trajectories therefore typically proceed by a **cascade of eliminations**, $W \to W-1 \to \cdots \to 1$, each one a symmetry-breaking event with its own timescale. Nothing in the $W = 2$ phenomenology prepares you for this, and it should be visible in the empowerment profile as structure at each elimination.

4 Symmetry, and how the LLM breaks it

4.1 The two group actions

Two distinct symmetries, and conflating them causes real errors:

$$\text{names: } \sigma \in S_W,\ (P_\sigma s)_i = \sigma(a_i) \qquad\qquad \text{agents: } \tau \in S_N,\ (Q_\tau s)_i = a_{\tau^{-1}(i)}.$$

S_N-equivariance is what licenses reduction to compositions. S_W-equivariance is what makes all names interchangeable. Our model satisfies both exactly:

$$P_\sigma\, \mathcal{T}_u\, P_\sigma^{-1} = \mathcal{T}_{\sigma(u)}, \qquad Q_\tau\, \mathcal{T}_u\, Q_\tau^{-1} = \mathcal{T}_u.$$

4.2 The quotient, and the loss of a complete invariant

S_W acts on the simplex by permuting barycentric coordinates. A fundamental domain is the sorted region $\{x_1 \geq x_2 \geq \cdots \geq x_W\}$, of relative volume $1/W!$, and the orbits are the partitions of N into at most W parts:

W	**compositions** ($N = 24$)	**orbits** $p_{\le W}(24)$	**max-share complete?**
2	25	13	**yes**
3	325	61	no
4	2 925	169	no

At $W = 2$ the orbit of $(n, N-n)$ is $\{(n, N-n), (N-n, n)\}$, and $\max(n, N-n)/N$ separates orbits completely: max-share *is* the quotient coordinate, and no information is lost by using it.

At $W \ge 3$ this fails. With $N = 24$, the compositions $(10, 7, 7)$ and $(10, 9, 5)$ lie in different orbits yet share a max-share of $10/24$.

> **At $W = 2$ a scalar order parameter is lossless; at $W \ge 3$ no scalar can be.** The quotient is still $(W-1)$-dimensional, so any single number — max share, Simpson index, name entropy — discards genuine structure. This is a new source of coarse-graining error with no analogue in the binary game, and it sits upstream of every estimate in §7.

4.3 Where the LLM breaks S_W — with their numbers

Our kernel is S_W-equivariant by construction. The LLM's is not, and the paper measures it. For $W = 2$ with names $\{Q, M\}$ they report

$$p(M \mid \{1{:}M, Q;\ 2{:}Q, M\}) = 0.848 \qquad \text{vs.} \qquad p(Q \mid \{1{:}Q, M;\ 2{:}M, Q\}) = 0.451,$$

two memory states exchanged by the relabelling $\sigma : Q \leftrightarrow M$. Equivariance demands equality. Hence

$$P_\sigma\, \mathcal{T}^{\mathrm{LLM}}\, P_\sigma^{-1} \ne \mathcal{T}^{\mathrm{LLM}}.$$

Define the symmetrisation and the *equivariance defect*

> $$\mathcal{T}^{\mathrm{sym}} = \frac{1}{W!} \sum_{\sigma \in S_W} P_\sigma^{-1}\, \mathcal{T}\, P_\sigma, \qquad \Delta = \mathcal{T} - \mathcal{T}^{\mathrm{sym}}.$$

$\mathcal{T}^{\mathrm{sym}}$ is the projection onto the S_W-invariant part; Δ is everything else. It is not a nuisance term — it is the mechanism of the collective bias they report. Our model has $\Delta = 0$ identically and therefore **cannot** produce collective bias.

4.4 Why the defect is richer at $W > 2$

At $W = 2$, S_2 has one non-trivial element and Δ collapses to a single number: which of two names is "strong". At general W, decompose the space of name-marginals into isotypic components of S_W: the trivial representation (the uniform part) and the *standard representation*, of dimension $W - 1$. The bias

$$b \in \mathbb{R}^W, \qquad \sum_c b_c = 0$$

lives in the standard representation and has $W - 1$ independent components.

> Collective bias at W names is a **vector**, not a sign. It specifies a full ranking of names, not merely a winner — consistent with the paper's finding that some names are systematically favoured and that the preferred names differ between models.

5 Dynamical consequences of $W > 2$

Control bandwidth grows. With a controller able to force any of W names, $H(U) \le \log_2 W$, so the empowerment ceiling rises from 1 bit at $W = 2$ to 2 bits at $W = 4$. More options means more room for a real signal.

Empowerment becomes directional. At $W = 2$, control authority is a scalar function of position on the segment — high in the middle, zero at the ends. At $W \ge 3$ it depends on position *and* direction: near an edge of the simplex, where a name is already extinct, pushing toward that extinct name has almost no effect, while pushing toward a surviving one still does. Authority is anisotropic, and a single scalar $\mathcal{E}_h$ averages that anisotropy away.

Convergence slows and acquires stages. The elimination cascade of §3.2 means consensus time is a sum of stage durations rather than a single first-passage time. Aggregating across episodes by round index therefore mixes episodes that are at different *stages*, not merely at different points of one coordinate — a sharper version of the ragged-length problem.

> **Consequence for structural zeros.** Under an S_W-invariant swept condition $(\varepsilon, \kappa, N, \lambda, k)$ every name is equally likely to win, so $I(C; \text{winner}) = 0$ exactly, for any W. But when the condition *is* the control input u, the symmetry is deliberately broken and $I(U; \text{winner}) > 0$ up to $\log_2 W$. These are a null and a positive control respectively, and they must not be conflated.

6 The irreducible differences

Three gaps that no amount of parameter matching closes.

1. Hidden state. Their observable macrostate is a function of unobserved memory buffers (§2.4). Conditioning on M_t never screens off the past, so a conditional MI computed on their data is a conditional MI *of a hidden Markov process*, carrying residual dependence through an unobservable confounder. Our action profile is the state, so conditioning is sufficient by construction.

2. Timing. They update one pair per step; we update all N. Their population round is $N/2$ interactions, so a horizon h is not comparable between the models without rescaling. The asynchronous rule also has different fluctuation structure — $O(1)$ change per step rather than $O(\sqrt{N})$ — so the two agree only in the mean-field limit, not in the noise.

3. Symmetry. $\Delta = 0$ for us, $\Delta \neq 0$ for them (§4.3). Our model can reproduce convergence to consensus, tipping points under a committed minority, and control authority — but not collective bias, unless a known asymmetry is deliberately inserted. Which, incidentally, is the only way to *validate* a bias detector: against an asymmetry whose size you set.

7 Statistics: estimating the MI when $W > 2$

Everything above is exact. This section is about what survives contact with finite samples.

7.1 The estimand must be a binned scalar

Three candidate macrostates, with the CMI table degrees of freedom $\text{dof} = (|\mathcal{U}|-1)(|\mathcal{M}|-1)|\mathcal{M}|$ and $|\mathcal{U}| = W$:

Macrostate	$\lvert M\rvert$ at $W = 4, N = 24$	dof	verdict
full composition	2 925	$\sim 2.6 \times 10^7$	hopeless
sorted composition	169	85 176	hopeless
binned scalar, $B = 3$	3	18	usable

So the estimand is forced: a scalar S_W-invariant, binned to a few levels. §4.2 already told us this is strictly lossy at $W \geq 3$. The exact computation may and should use the full composition; the *estimator* cannot. Ground truth and estimate must then be compared on the binned representation — never on different ones.

7.2 Bias grows faster than signal

Plug-in bias is $\mathrm{dof}/(2n_{\mathrm{eff}} \ln 2)$ with n_{eff} the number of *episodes*. The signal ceiling grows like $\log_2 W$; the dof grow like $(W-1)B(B-1)$. Taking the walkthrough's worked value of 0.07 bits at $W = 2$ and scaling the signal optimistically as $\log_2 W$:

W	B	dof	plausible true MI	episodes for 10% bias
2	3	6	0.07	~620
3	3	12	0.11	~780
4	3	18	0.14	~930
4	5	60	0.14	~3 100

> **Binning dominates W.** Going from two names to four costs about 1.5× the episodes. Going from three bins to five, at fixed $W = 4$, costs about 3.3×. The number of bins is the expensive decision, not the size of the name pool.

7.3 Three traps specific to $W > 2$

Bin edges must depend on W. The max share ranges over $[1/W, 1]$, not $[0, 1]$. Fixed $[0, 1]$ edges leave the lowest bins structurally empty at $W = 4$ — dof spent on cells that can never be occupied, so pure bias with no possibility of signal.

Never sweep W itself against a share-based macrostate. The support of the share differs with W, so the condition becomes recoverable from a single macrostate observation and the MI inflates toward $H(C)$ for reasons of alphabet support alone. If W must be a swept condition, bin onto a grid common to all its values.

Nulls do not transfer across W. The shuffle null's location and width are set by the dof, which change with W and B. A null distribution computed at $W = 2$ says nothing about significance at $W = 4$. Recompute per configuration.

7.4 What to report

Given that a scalar estimand is forced and lossy, the honest reporting is:

1. the exact value on the full composition, computed from the kernel;
2. the exact value on the binned scalar — the gap between the two *is* the coarse-graining cost, and it is computable, not a matter of opinion;

3. the estimate, with the plug-in bias for its dof stated alongside;
4. the null band for that exact (W, B, n_{eff}).

Item 2 is the one that does not exist at $W = 2$ and is easy to forget. Without it, a difference between exact and estimated values at $W = 4$ cannot be attributed between coarse-graining and sampling.